

SOME MIXTURES OF THE VON MISES DISTRIBUTION, II

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SUMMARY

In this paper we find three mixtures of von Mises distribution using a Von Mises distribution with different constants, a wrapped normal and a symmetric triangular distribution as distributions of the von Mises location parameter. All of these mixtures appear to be new circular distributions.

1. Introduction

Let $T = \{F\}$ be a family of distributions and μ a measure on a Borel field of subsets of T with $\mu(T) = 1$. Then $H(\cdot) = \int_T F(\cdot) d\mu(F)$ is again a distribution function called a μ -mixture of T . Let $f(\cdot, \theta)$ be the probability density function (p.d.f.) of a random variable A , where the range of A is the points on the circumference of a circle. If the parameter θ is itself a random variable with the p.d.f. $g(\cdot, \beta)$, where β is its parameter, then the g -mixture of f is defined by $\int_{\theta} f \cdot g \, d\theta$.

The von Mises circular distribution plays a prominent role in statistical inference on the circle and its importance there is almost the same as that of the normal distribution on the line. See Mardia (1972). In this paper we have obtained mixtures of this distribution using a von Mises distribution with different constants, a wrapped normal circular distribution and a symmetric triangular circular distribution as distributions of the location parameter. All of these mixtures seem to be new circular distributions.

Here we skip the discussion of the probability space Ω , the distribution function $F(\alpha)$, the p.d.f. $f(\alpha)$ of the random variable A and its trigonometric moments as they have been discussed by Gupta (1976).

2. Some distributions

(1) *The von Mises circular distribution* has the p.d.f.

$$(2.1) \quad f(\alpha) = \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)}, \quad \alpha, \alpha_0 \in (0, 2\pi], k \geq 0.$$

where $I_\nu(k)$, for any integer ν , denotes the ν -th Bessel function with purely imaginary argument, and k and α_0 are the parameters. Its trigonometric moments are

$$\Phi_f(0) = 1 \text{ and } \Phi_f(\nu) = \frac{I_\nu(k)}{I_0(k)} \exp(i\nu\alpha_0), \nu \neq 0$$

In this paper, we assume that k is fixed.

(2) *The wrapped normal circular distribution* was introduced by Zernike (1928) and later studied by Wintner (1933), Levy (1939), Gumbel, Greenwood, and Durand (1953). It is generated by wrapping the ordinary normal distribution with p.d.f. $n(0, \sigma^2)$ around a circle. Its p.d.f. is given as

$$(2.2) \quad f(\alpha) = 1/\pi [1/2 + \sum_{r=1}^{\infty} \rho r^2 \cos r(\alpha - \theta)], \quad \alpha, \theta \in (0, 2\pi], \rho \in [0, 1],$$

where $\rho = \exp(-\frac{1}{2}\sigma^2)$. Its trigonometric moments are given by

$$\Phi_f(0) = 1, \Phi_f(\nu) = \rho \nu^2 \exp(i\nu\theta), \nu \neq 0.$$

(3) *The symmetric triangular distribution* has p.d.f. given by

$$(2.3) \quad f(\alpha) = 1/\pi [1/2 + \rho \sum_{r=1}^{\infty} \frac{\cos(2r-1)(\alpha-\theta)}{(2r-1)^2}], \quad \alpha, \theta \in (0, 2\pi], \rho \in [0, 4/\pi^2].$$

Its trigonometric moments are

$$\Phi_f(0) = 1, \Phi_f(\nu) = \rho/\nu^2 \exp(i\nu\theta) \text{ if } \nu \text{ is odd and } \Phi_f(\nu) = 0 \text{ if } \nu \text{ is even, } \nu \neq 0.$$

3. Mixtures of the Von Mises circular distribution

The g-mixture of the p.d.f. $f(\alpha, \theta)$ from the introduction is

$$(3.1) \quad h(\alpha, \beta) = \int_{\theta} f(\alpha, \theta) g(\theta, \beta) d\theta.$$

The lemma 1 of Gupta (1976) will be used in this paper. For convenience we repeat it.

Lemma 1. Let $\Phi_f(\nu, \theta)$ and $\Phi_h(\nu)$ denote the ν -th trigonometric moments, respectively, of

the density functions $f(\alpha, \theta)$ and $h(\alpha, \beta)$. Then

$$(3.2) \quad \Phi_h(\nu) = \int_{\theta} \Phi_f(\nu, \theta) g(\theta, \beta) d\theta.$$

We now give three mixtures of the Von Mises distribution.

3.1 Mixing with the Von Mises distribution

Let

$$(3.3) \quad f(\alpha, \alpha_0) = \frac{\exp(k_1 \cos(\alpha - \alpha_0))}{2\pi I_0(k_1)}, \quad \alpha, \alpha_0 \in (0, 2\pi], \quad k_1 > 0,$$

where k_1 is fixed and let α_0 have another Von Mises distribution having the 'mixing' density function

$$(3.4) \quad g(\alpha_0; k_2, \beta) = \frac{\exp(k_2 \cos(\alpha_0 - \beta))}{2\pi I_0(k_2)}, \quad \alpha_0, \beta \in (0, 2\pi], \quad k_2 > 0,$$

k_2 and β being known parameters. The g-mixture of the density function $f(\alpha, \alpha_0)$ is

$$(3.5) \quad h(\alpha; k_1, k_2, \beta) = \int_0^{2\pi} \frac{\exp(k_1 \cos(\alpha - \alpha_0) + k_2 \cos(\alpha_0 - \beta))}{2\pi I_0(k_1) \cdot 2\pi I_0(k_2)} d\alpha_0.$$

$$\text{Now } k_1 \cos(\alpha - \alpha_0) + k_2 \cos(\alpha_0 - \beta) = \sqrt{k_1^2 + k_2^2} \sqrt{p^2 + q^2} \cos(\alpha_0 - \gamma).$$

where $\tan \delta = k_1/k_2$, $p = \sin \delta \cos \alpha + \cos \delta \cos \beta$, $q = \sin \delta \sin \alpha + \cos \delta \sin \beta$ and $\tan \gamma = q/p$. Therefore,

$$\begin{aligned} (3.6) \quad h(\alpha; k_1, k_2, \beta) &= \int_0^{2\pi} \frac{\exp(\sqrt{(k_1^2 + k_2^2)(p^2 + q^2)} \cos(\alpha_0 - \gamma))}{2\pi I_0(k_1) \cdot 2\pi I_0(k_2)} d\alpha_0 \\ &= \frac{I_0(\sqrt{(k_1^2 + k_2^2)(p^2 + q^2)})}{2\pi I_0(k_1) I_0(k_2)} \\ &= \frac{I_0(\sqrt{(k_1^2 + k_2^2)(1 + \sin 2\delta \cos(\alpha - \beta))})}{2\pi I_0(k_1) I_0(k_2)}, \quad k_1, k_2 > 0, \alpha, \beta \in (0, 2\pi], \end{aligned}$$

on expanding $p^2 + q^2$ under the root sign. This seems to be a new circular distribution.

We call it the Bessel function circular distribution. By using lemma 1, the trigonometric moments of this g-mixture are obtained as

$$\begin{aligned}
 (3.7) \quad \Phi_h(\nu) &= \int_0^{2\pi} \frac{I_\nu(k_1) \exp(i\nu\alpha_0) \exp(k_2 \cos(\alpha_0 - \beta))}{I_0(k_1) 2\pi I_0(k_2)} d\alpha_0, \nu \neq 0 \\
 &= \frac{I_\nu(k_1)}{I_0(k_1)} \exp(i\nu\beta) \int_0^{2\pi} \frac{\exp(i\nu(\alpha_0 - \beta) + k_2 \cos(\alpha_0 - \beta))}{2\pi I_0(k_2)} d\alpha_0 \\
 &= \frac{I_\nu(k_1)}{I_0(k_1)} \exp(i\nu\beta) \frac{I_\nu(k_2)}{I_0(k_2)} = \exp(i\nu\beta) \prod_{j=1}^2 \frac{I_\nu(k_j)}{I_0(k_j)}.
 \end{aligned}$$

And $\Phi_h(0) = 1$.

3.2 Mixing with wrapped normal distribution

Consider the Von Mises density function (2.1) where k is fixed and α_0 has the density function

$$(3.8) \quad g(\alpha_0, \beta) = 1/\pi [1/2 + \sum_{r=1}^{\infty} \rho r^2 \cos r(\alpha_0 - \beta)], \beta, \alpha_0 \in (0, 2\pi], \rho \in [0, 1],$$

$$\text{where } \rho = \exp(-\sigma^2/2), \sigma^2 > 0.$$

The g-mixture of the density function $f(\alpha, \alpha_0)$ is

$$\begin{aligned}
 (3.9) \quad h(\alpha, \beta) &= \int_0^{2\pi} \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} \cdot 1/\pi [1/2 + \sum_{r=1}^{\infty} \rho r^2 \cos r(\alpha_0 - \beta)] d\alpha_0 \\
 &= 1/2\pi + 1/\pi \int_0^{2\pi} \sum_{r=1}^{\infty} \frac{\rho r^2 \exp(k \cos(\alpha - \alpha_0)) \cos r(\alpha_0 - \beta)}{2\pi I_0(k)} d\alpha_0.
 \end{aligned}$$

Since the infinite series under the integral sign converges for every value of $\rho, \rho \in (0, 1)$, we may integrate term by term obtaining $h(\alpha, \beta)$ as

$$\begin{aligned}
 h(\alpha, \beta) &= 1/2\pi + 1/\pi \sum_{r=1}^{\infty} \rho r^2 \operatorname{Re} \left[\int_0^{2\pi} \frac{\exp(k \cos(\alpha - \alpha_0) + ir(\alpha_0 - \beta))}{2\pi I_0(k)} d\alpha_0 \right] \\
 &= 1/2\pi + 1/\pi \sum_{r=1}^{\infty} \rho r^2 \operatorname{Re} [\exp(ir(\alpha - \beta)) \int_0^{2\pi} \frac{\exp(k \cos(\alpha_0 - \alpha) + ir(\alpha_0 - \alpha))}{2\pi I_0(k)} d\alpha_0].
 \end{aligned}$$

The integral on the right equals the r -th trigonometric moment $\Phi_r(r)$ of the random angle A_0 having the Von Mises distribution with fixed α . Hence

$$(3.10) \quad h(\alpha, \beta) = 1/2\pi + 1/\pi \sum_{r=1}^{\infty} \rho r^2 R_1[\exp(i r(\alpha - \beta)) \frac{I_r(k)}{I_0(k)}] \\ = 1/2\pi + 1/\pi \sum_{r=1}^{\infty} \rho r^2 \frac{I_r(k)}{I_0(k)} \cos r(\alpha - \beta).$$

This distribution also seems to be a new circular distribution with trigonometric moments

$$(3.11) \quad \Phi_h(\nu) = 1, \nu = 0 \text{ and } \Phi_h(\nu) = \rho \nu^2 \frac{I_\nu(k)}{I_0(k)} \exp(i\nu\beta), \nu \neq 0$$

3.3 Mixing with a symmetric triangular distribution

As before consider the von Mises density function (2.1) where k is fixed and α_0 has the density function

$$(3.12) \quad g(\alpha_0, \beta) = 1/\pi [1/2 + \rho \sum_{r=1}^{\infty} \frac{\cos(2r-1)(\alpha_0 - \beta)}{(2r-1)^2}], \quad \alpha_0, \beta \in (0, 2\pi], \rho \in (0, 4/\pi^2].$$

The g -mixture in this case is

$$(3.13) \quad h(\alpha, \beta) = \int_0^{2\pi} \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} \cdot \frac{1}{\pi} [1/2 + \rho \sum_{r=1}^{\infty} \frac{\cos(2r-1)(\alpha_0 - \beta)}{(2r-1)^2}] d\alpha_0$$

The above integral can be evaluated in the same way as in subsection 3.2. It leads to

$$(3.14) \quad h(\alpha, \beta) = 1/\pi [1/2 + \rho \sum_{\nu=1}^{\infty} \frac{I_{2\nu-1}(k)}{I_0(k)} \cdot \frac{\cos(2\nu-1)(\alpha - \beta)}{(2\nu-1)^2}]$$

This also seems to be a new circular distribution with trigonometric moments

$$(3.15) \quad \Phi_h(0) = 1, \text{ if } \nu = 0; \Phi_h(\nu) = 0, \text{ if } \nu \text{ is even and } \nu \neq 0;$$

$$\Phi_h(\nu) = \rho/\nu^2 \cdot \frac{I_\nu(k)}{I_0(k)} \exp(i\nu\beta), \text{ if } \nu \text{ is odd.}$$

REFERENCES

- GUPTA, D. (1976). Some Mixtures of Von Mises Distributions, I, *South African Statist. J.* 10 69-75.
GUMBEL, E.J., GREENWOOD, J.A., and DURAND, D. (1953). The Circular Normal

- Distribution: Theory and Tables. *Jour. Amer. Statist. Assoc.* 48 131-152.
- LEVY, P. (1939). L'addition Des Variables Aléatoires Définies Sur Une Circonférence. *Bull. Soc. Math. France* 67 1-41.
- MARDIA, K.V. (1972). *Statistics of Directional Data*. Academic Press, New York.
- WINTNER, A. (1933). On the Stable Distribution Law. *Amer. Jour. of Math.* 55 335-339.
- ZERNIKE, F. (1928). Wahrscheinlichkeitsrechnung und Mathematische Statistik. In *"Handbuch der Physik"* 3 419-92. Springer, Berlin.

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